

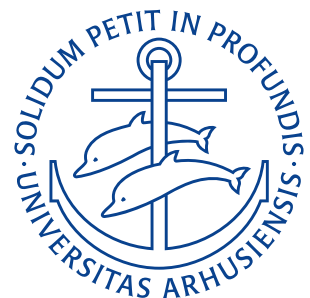
**Department of Mechanical and Production Engineering**

Aarhus University

# Exercises

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**Lecture 1: Fourier Theory**

29. August 2025

**Problem 1.1**

Consider the following periodic functions. Find possible values for the period, using the definition of periodic functions.

- a)  $f(x) = e^{\cos x}$
- b)  $f(x) = \cos\left(x + \frac{\pi}{3}\right)$
- c)  $f(x) = \cos\left(\frac{\pi x}{3}\right)$
- d)  $f(x) = \cos x + \cos(3x)$
- e)  $f(x) = \cos(x) \cos(3x)$
- f)  $f(x) = \cos x + \cos(0,6x)$

**Derivation**

- a) From the lecture (p. 1), we know that a function  $f$  is periodic, with period  $p$ , if

$$f(x) = f(x + p).$$

Using this definition we obtain:

$$\begin{aligned} e^{\cos x} &= e^{\cos(x+p)} \\ \cos x &= \cos(x + p) \\ \Rightarrow p &= 2\pi, 4\pi, 6\pi, \dots \end{aligned}$$

- b) We use the same definition of periodicity to obtain

$$\begin{aligned} \cos\left(x + \frac{\pi}{3}\right) &= \cos\left(x + p + \frac{\pi}{3}\right) \\ \Rightarrow p &= 2\pi, 4\pi, 6\pi. \end{aligned}$$

- c) Following the same procedure we get

$$\begin{aligned} \cos\left(\frac{\pi x}{3}\right) &= \cos\left(\frac{\pi(x+p)}{3}\right) \\ \cos\left(\frac{\pi x}{3}\right) &= \cos\left(\frac{\pi x}{3} + \frac{\pi p}{3}\right) \\ \Rightarrow \frac{\pi p}{3} &= 2\pi, 4\pi, 6\pi, \dots \\ \Rightarrow p &= 6, 12, 18, \dots \end{aligned}$$

- d) We follow the same procedure and get

$$\begin{aligned} \cos x + \cos(3x) &= \cos(x + p) + \cos(3(x + p)) \\ \cos x + \cos(3x) &= \cos(x + p) + \cos(3x + 3p). \end{aligned}$$

This holds if  $p$  is an integer multiple of  $2\pi$  and  $3p$  is an integer multiple of  $2\pi$ , i.e.  $p = 2\pi, 4\pi, 6\pi, \dots$

- e) Once again, we follow the procedure and obtain

$$\begin{aligned} \cos(x) \cos(3x) &= \cos(x + p) \cos(3(x + p)) \\ \cos(x) \cos(3x) &= \cos(x + p) \cos(3x + 3p). \end{aligned}$$

From the above, we see that  $p = \pi, 2\pi, 3\pi$ . This is due to the fact, that the two functions are multiplied, hence  $\cos(0) \cdot \cos(0) = \cos(\pi) \cos(3\pi) = 1$

f) We follow the same procedure as in the previous subproblems and obtain

$$\cos x + \cos(0,6x) = \cos(x + p) + \cos(0,6x + 0,6p).$$

I.e.  $p$  must be an integer multiple of  $2\pi$  and  $0,6p$  must be an integer multiple of  $2\pi$ . Hence  $p = 10\pi, 20\pi, 30\pi, \dots$

- a)  $p = 2\pi, 4\pi, 6\pi, \dots$
- b)  $p = 2\pi, 4\pi, 6\pi, \dots$
- c)  $p = 6, 12, 18, \dots$
- d)  $p = 2\pi, 4\pi, 6\pi, \dots$
- e)  $p = \pi, 2\pi, 3\pi, \dots$
- f)  $p = 10\pi, 20\pi, 30\pi, \dots$

RESULT

## Problem 1.2

Let

$$f(x) = x, \quad -1 \leq x \leq 1, \quad f(x) = f(x + p), \quad p = 2.$$

- Sketch  $f$
- Find a Fourier series representation of  $f$ .

### Derivation

- On [Figure 0.1](#) the periodic repeating function  $f(x)$  is depicted.

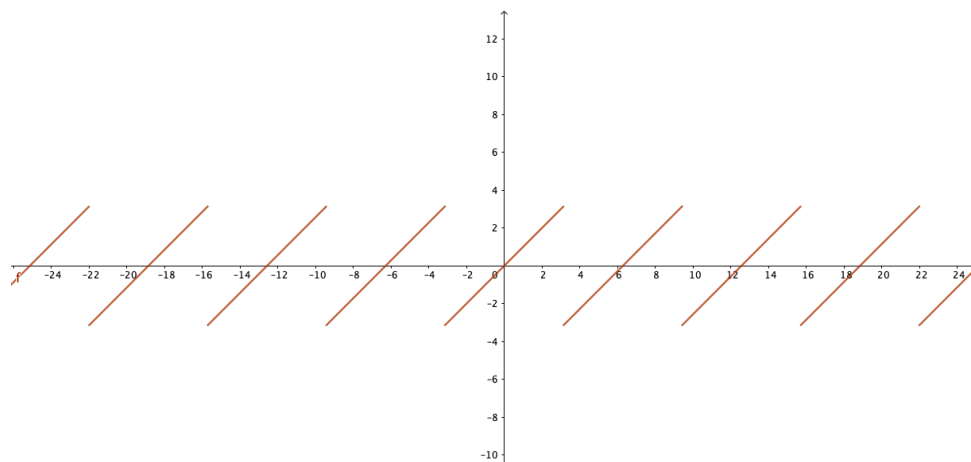


Figure 0.1: Sketch of  $f$ .

- From the lecture (p. 5), we have the general representation of a Fourier series as

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left( a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right)$$

for

$$a_0 = \frac{1}{2L} \int_{-L}^L f(x) \, dx$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) \, dx, \quad n = 1, 2, 3, \dots$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) \, dx, \quad n = 1, 2, 3, \dots$$

In this case,  $L = \frac{p}{2} \implies L = 1$ . Therefore:

$$a_0 = \frac{1}{2} \int_{-1}^1 x \, dx$$

$$= \frac{1}{2} \left[ \frac{x^2}{2} \right]_{-1}^1$$

$$= \frac{1}{2} \left( \frac{1}{2} - \frac{1}{2} \right)$$

$$= 0$$

$$a_n = \int_{-1}^1 x \cos(n\pi x) \, dx$$

$$\begin{aligned}
&= 0 \\
b_n &= \int_{-1}^1 x \sin(n\pi x) \, dx \\
&= \left[ \frac{\sin(\pi n x)}{\pi^2 n^2} - \frac{x \cos(\pi n x)}{\pi n} \right]_{-1}^1 \\
&= \left[ -\frac{x \cos(\pi n x)}{\pi n} \right]_{-1}^1 \\
&= \left( -\frac{\cos(n\pi)}{\pi n} + \left( \frac{\cos(-n\pi)}{\pi n} \right) \right) \\
&= \left( -\frac{\cos(n\pi)}{\pi n} - \frac{\cos(n\pi)}{\pi n} \right) \\
&= -\frac{2}{\pi n} \cos(\pi n).
\end{aligned}$$

Hence, the complete Fourier representation is

$$f(x) = \sum_{n=1}^{\infty} -\frac{2}{\pi n} \cos(\pi n) \sin(n\pi x).$$

a) See Figure 0.1

b) 
$$f(x) = \sum_{n=1}^{\infty} -\frac{2}{\pi n} \cos(\pi n) \sin(n\pi x).$$

RESULT

**Problem 1.3**

Let  $a \neq 0$  and  $b \neq 0$ . Let  $f(x) = f(x + p_0)$  be a periodic function with period  $p_0$ . Is  $f(ax + b)$  periodic? If yes, find a period. Give arguments for your answer.

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**Derivation**

We let  $f(x) = f(x + p)$  denote the function given in the assignment and  $g(x) = f(ax + b)$  be the function which we wish to determine is periodic or not. To determine if  $g(x)$  is periodic we must determine if  $g(x) = g(x + p)$  holds. We get:

$$\begin{aligned} g(x + p) &= g(x) \\ &= f(a(x + p) + b) \\ &= f(ax + b + ap). \end{aligned}$$

Comparing this with the original function  $f$  we see that it repeats when  $ap = p_0$ . Since  $a \neq 0$  we can divide by  $a$  and obtain

$$p = \frac{p_0}{a}.$$

The function is periodic with period  $p = \frac{p_0}{a}$

RESULT

**Problem 1.4**

Calculate the left-hand limit and the right hand limit of

$$f(x) = \frac{|x|}{x}$$

at  $x = 0$ .

---

**Derivation**

From the lecture (p. 2), we know that the left hand limit of  $f(x)$  at  $x_0$  is defined as

$$f(x_0-) = \lim_{h \rightarrow 0} f(x_0 - h)$$

and the right hand limit of  $f(x)$  at  $x_0$  is

$$f(x_0+) = \lim_{h \rightarrow 0} f(x_0 + h)$$

for positive  $h$ .

Using these two definitions we get:

$$\begin{aligned} f(0-) &= \lim_{h \rightarrow 0} f(0 - h) \\ &= \lim_{h \rightarrow 0} \frac{|-h|}{-h} \\ &= \lim_{h \rightarrow 0} \frac{|h|}{-h} \\ &= -1 \\ f(0+) &= \lim_{h \rightarrow 0} f(0 + h) \\ &= \lim_{h \rightarrow 0} \frac{|h|}{h} \\ &= 1. \end{aligned}$$

|                                                            |
|------------------------------------------------------------|
| $\begin{aligned} f(0-) &= -1 \\ f(0+) &= 1. \end{aligned}$ |
|------------------------------------------------------------|

RESULT

## Problem 1.5

Calculate the left-hand derivative and the right-hand derivative of

$$f(x) = x|x|$$

at  $x = 0$ .

### Derivation

From the lecture (p. 4) the definition of the left hand derivative of  $f(x)$  at  $x_0$  is defined as

$$f'(x_0-) = \lim_{h \rightarrow 0} \frac{f(x_0) - f(x_0 - h)}{h}$$

and the right hand derivative of  $f(x)$  at  $x_0$  is defined as

$$f'(x_0+) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

for positive  $h$ .

Using these definitions we get:

$$\begin{aligned} f'(0-) &= \lim_{h \rightarrow 0} \frac{f(0) - f(-h)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-(-h|-h|)}{h} \\ &= \lim_{h \rightarrow 0} \frac{h|h|}{h} \\ &= \lim_{h \rightarrow 0} |h| \\ &= 0 \\ f'(0+) &= \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{h|h|}{h} \\ &= \lim_{h \rightarrow 0} |h| \\ &= 0. \end{aligned}$$

|                                                             |
|-------------------------------------------------------------|
| $\begin{aligned} f'(0-) &= 0 \\ f'(0+) &= 0. \end{aligned}$ |
|-------------------------------------------------------------|

RESULT

## Lecture 2: Even and odd functions

5. September 2025

### Problem 2.1

Find a period and the corresponding Fourier coefficients of

- a)  $f(x) = \sin\left(\frac{\pi}{4}\right) \sin\left(\frac{2x}{4}\right) + \cos\left(\frac{\pi}{5}\right) \sin\left(\frac{3x}{7}\right)$
- b)  $f(x) = \sin(\sqrt{2}x) + \cos\left(\frac{x}{\sqrt{2}}\right)$
- c)  $f(x) = \sin\left(\frac{\pi}{3} + \frac{3\pi x}{4}\right)$ .

*Hint:* You may want to use

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \sin \beta \cos \alpha.$$

### Derivation

- a) We start by rewriting the given function as

$$f(x) = \sin\left(\frac{\pi}{4}\right) \sin\left(\frac{7\pi x}{14\pi}\right) + \cos\left(\frac{\pi}{5}\right) \sin\left(\frac{6\pi x}{14\pi}\right).$$

Now, comparing with the general form of a Fourier series:

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left( a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right)$$

We see that  $L = 14\pi$ ,  $p = 28\pi$ ,  $b_6 = \cos(\pi/5)$  and  $b_7 = \sin(\pi/4)$ . All other Fourier coefficients are 0.

- b) We rewrite as:

$$f(x) = \sin\left(\frac{2\pi x}{\sqrt{2}\pi}\right) + \cos\left(\frac{\pi x}{\sqrt{2}\pi}\right).$$

By comparison with the general form we see that  $L = \sqrt{2}\pi$ ,  $p = 2\sqrt{2}\pi$ ,  $b_2 = 1$ , and  $a_1 = 1$ . All other Fourier coefficients are 0.

- c) We use the hint to rewrite as

$$\begin{aligned} f(x) &= \sin\left(\frac{\pi}{3} + \frac{3\pi x}{4}\right) \\ &= \sin\left(\frac{\pi}{3}\right) \cos\left(\frac{3\pi x}{4}\right) + \cos\left(\frac{\pi}{3}\right) \sin\left(\frac{3\pi x}{4}\right). \end{aligned}$$

By comparison, we see that  $L = 4$ ,  $p = 8$ ,  $a_3 = \sin(\pi/3)$ , and  $b_3 = \cos(\pi/3)$ . All other Fourier coefficients are zero

|    |                     |                                         |                                         |
|----|---------------------|-----------------------------------------|-----------------------------------------|
| a) | $p = 28\pi,$        | $b_6 = \cos\left(\frac{\pi}{5}\right),$ | $b_7 = \sin\left(\frac{\pi}{4}\right),$ |
| b) | $p = 2\sqrt{2}\pi,$ | $a_1 = 1,$                              | $b_2 = 1,$                              |
| c) | $p = 8,$            | $a_3 = \sin\left(\frac{\pi}{3}\right),$ | $b_3 = \cos\left(\frac{\pi}{3}\right).$ |

RESULT

## Problem 2.2

Are the following functions even or odd? Give arguments.

a)  $f(x) = (\cos(x^2) + \cos(x)) \sin(x)$

b)  $f(x) = \frac{1 + \cos x}{2 + \cos(x^2)}$

c)  $f(x) = \frac{1 + \sin(x)}{2 + \sin(x^2)}$

d)  $f(x) = x|x|$ .

## Derivation

From the lecture (p. 3), we know that a function  $f$  is even if

$$f(x) = f(-x) \quad \text{for all } x$$

and that a function  $f$  is odd if

$$f(x) = -f(-x) \quad \text{for all } x.$$

We will therefore try to insert  $-x$  in the place of  $x$  and see if that causes a change of sign for each of the given functions.

We will however, instead utilize the simple fact that multiplication of functions follow the same rules for even/odd as multiplication of numbers. Furthermore, we know that  $x$  is odd,  $\cos$  is even,  $\sin$  is odd and  $|x|$  is even. In the following, odd functions are called  $O$  and even functions are called  $E$ .

a) Using the method described above we get:

$$\begin{aligned} f(x) &= (\cos(x \cdot x) + \cos(x)) \sin(x) \\ &= \cos(x \cdot x) \sin x + \cos(x) \sin(x) \\ &= E(O \cdot O) \cdot O + E \cdot O \\ &= E \cdot O + E \cdot O \\ &= O + O \\ &= O. \end{aligned}$$

b) Again,

$$\begin{aligned} f(x) &= \frac{1 + \cos x}{2 + \cos(x \cdot x)} \\ &= \frac{E + E}{E + E} \\ &= \frac{E}{E} \\ &= E. \end{aligned}$$

c) Again,

$$\begin{aligned} f(x) &= \frac{1 + \sin x}{2 + \sin(x \cdot x)} \\ &= \frac{E + O}{E + O(E)} \\ &= \frac{E + O}{E + E} \end{aligned}$$

$$\begin{aligned} &= \frac{E + O}{E} \\ &= \frac{E}{E} + \frac{O}{E} \\ &= E + O. \end{aligned}$$

As the sum of an even and an odd function is neither even nor odd if none of the functions are 0. Hence the function in question is neither even nor odd.

d) Lastly,

$$\begin{aligned} f(x) &= x \cdot |x| \\ &= O \cdot E \\ &= O. \end{aligned}$$

- a) Odd,
- b) Even,
- c) Neither even nor odd,
- d) Odd.

RESULT

### Problem 2.3

Consider the periodic function

$$f(x) = \begin{cases} x, & -2 < x < -1 \\ x + k, & -1 < x < 1 \\ x, & 1 < x < 2 \end{cases}$$

$$f(x) = f(x + p), p = 4$$

where  $k$  is a constant. Decompose  $f(x)$  into its even and odd part.

#### Derivation

For  $-2 < x < -1$  we get

$$f_1(x) = \frac{1}{2} (f(x) + f(-x)) = \frac{1}{2} (x + (-x)) = 0$$

$$f_2(x) = \frac{1}{2} (f(x) - f(-x)) = \frac{1}{2} (x - (-x)) = x.$$

For  $-1 < x < 1$  we get

$$f_1(x) = \frac{1}{2} (f(x) + f(-x)) = \frac{1}{2} (x + k + (-x + k)) = k$$

$$f_2(x) = \frac{1}{2} (f(x) - f(-x)) = \frac{1}{2} (x + k - (-x + k)) = x.$$

And for  $1 < x < 2$  we get

$$f_1(x) = \frac{1}{2} (f(x) + f(-x)) = \frac{1}{2} (x + (-x)) = 0$$

$$f_2(x) = \frac{1}{2} (f(x) - f(-x)) = \frac{1}{2} (x - (-x)) = x.$$

We thus have the decomposed function  $f(x)$

$$f(x) = f_1(x) + f_2(x)$$

with

$$f_1(x) = \begin{cases} 0, & -2 < x < -1 \\ k, & -1 < x < 1 \\ 0, & 1 < x < 2. \end{cases}$$

$$f_1(x) = f_1(x + p), \quad p = 4$$

and

$$f_2(x) = x, \quad -2 < x < 2$$

$$f_2(x) = f_2(x + p), \quad p = 4.$$

$$f(x) = f_1(x) + f_2(x)$$

$$f_1(x) = \begin{cases} 0, & -2 < x < -1 \\ k, & -1 < x < 1 \\ 0, & 1 < x < 2. \end{cases}$$

$$f_1(x) = f_1(x+p), \quad p=4$$

$$f_2(x) = x, \quad -2 < x < 2$$

$$f_2(x) = f_2(x+p), \quad p=4.$$

RESULT

## Problem 2.4

Consider the periodic function

$$f(x) = \begin{cases} 0, & -\pi < x < 0 \\ 2x, & 0 < x < \pi \end{cases}$$

$$f(x) = f(x + p), p = 2\pi.$$

Decompose  $f(x)$  into its even and odd part and find its Fourier series.

Hint: You may want to use examples from the lecture.

### Derivation

For  $-\pi < x < 0$  we get

$$f_1(x) = \frac{1}{2} (f(x) + f(-x)) = \frac{1}{2} (0 + (-2x)) = -x$$

$$f_2(x) = \frac{1}{2} (f(x) - f(-x)) = \frac{1}{2} (0 - (-2x)) = x.$$

For  $0 < x < \pi$  we get

$$f_1(x) = \frac{1}{2} (f(x) + f(-x)) = \frac{1}{2} (2x + (0)) = x$$

$$f_2(x) = \frac{1}{2} (f(x) - f(-x)) = \frac{1}{2} (2x - (0)) = x.$$

Therefore

$$f_1(x) = |x|, -\pi < x < \pi.$$

From the lecture (p. 7) the Fourier cosine series of this is

$$f_1(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{\pi n^2} (\cos(n\pi) - 1) \cos(nx).$$

Also

$$f_2(x) = x, -\pi < x < \pi.$$

Which has the Fourier sine series (p. 6)

$$f_2(x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{2}{n} \sin(nx).$$

Combining these two Fourier series we get the complete Fourier series:

$$f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \left( \frac{2}{\pi n^2} (\cos(n\pi) - 1) \cos(nx) + (-1)^{n+1} \frac{2}{n} \sin(nx) \right).$$

$$f_1(x) = |x|, \quad -\pi < x < \pi$$

$$f_2(x) = x, \quad -\pi < x < \pi$$

$$f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \left( \frac{2}{\pi n^2} (\cos(n\pi) - 1) \cos(nx) + (-1)^{n+1} \frac{2}{n} \sin(nx) \right).$$

RESULT

## Problem 2.5

Decompose

$$f(x) = e^{ix}$$

into its even and odd part.

Hint; you may want to use the representation of  $f(x)$  in terms of  $\cos x$  and  $\sin x$ .

### Derivation

We rewrite the function as

$$f(x) = e^{ix} = \cos x + i \sin x.$$

Now, to decompose we follow the same procedure as above:

$$\begin{aligned} f_1(x) &= \frac{1}{2} (f(x) + f(-x)) \\ &= \frac{1}{2} (\cos x + i \sin x + \cos(-x) + i \sin(-x)) \\ &= \frac{1}{2} (\cos x + i \sin x + \cos x - i \sin(x)) \\ &= \cos x \\ f_2(x) &= \frac{1}{2} (f(x) - f(-x)) \\ &= \frac{1}{2} (\cos x + i \sin x - \cos(-x) - i \sin(-x)) \\ &= \frac{1}{2} (\cos x + i \sin x - \cos x + i \sin x) \\ &= i \sin x. \end{aligned}$$

|                                                                         |
|-------------------------------------------------------------------------|
| $\begin{aligned} f_1(x) &= \cos x \\ f_2(x) &= i \sin x. \end{aligned}$ |
|-------------------------------------------------------------------------|

RESULT

## Lecture 3: Fourier Series

12. September 2025

### Problem 3.1

Consider the function  $f(x) = \cos x, 0 < x < \pi$ .

- Sketch the expansion of  $f$  and the corresponding Fourier series.
- Sketch the even expansion of  $f$  and find the corresponding Fourier cosine series.
- Sketch the odd expansion of  $f$  and find the corresponding Fourier cosine series.
- Calculate the error  $E_N$  of the expansion and the odd expansion for  $N = 1, 2, 3$ . What can be said about the quality of the approximations for these two expansions=

Hint: In some cases the expansion can be an even/odd function.

Hint: For  $m, n \in \mathbb{N}, m \neq n, L > 0$ :

$$\int_0^{\frac{\pi}{2}} \cos x \sin(2nx) dx = -\frac{2n}{1-4n^2}$$

$$\int_0^L \cos\left(\frac{m\pi x}{L}\right) \sin\left(\frac{n\pi x}{L}\right) dx = \frac{nL}{\pi(m^2 - n^2)} (\cos(n\pi) \cos(m\pi) - 1)$$

$$\int_0^L \cos\left(\frac{\pi x}{L}\right) \sin\left(\frac{\pi x}{L}\right) dx = 0.$$

### Derivation

- The expansion of  $f$  is depicted on [Figure 0.2](#). Based on the figure the expansion looks to generate an odd

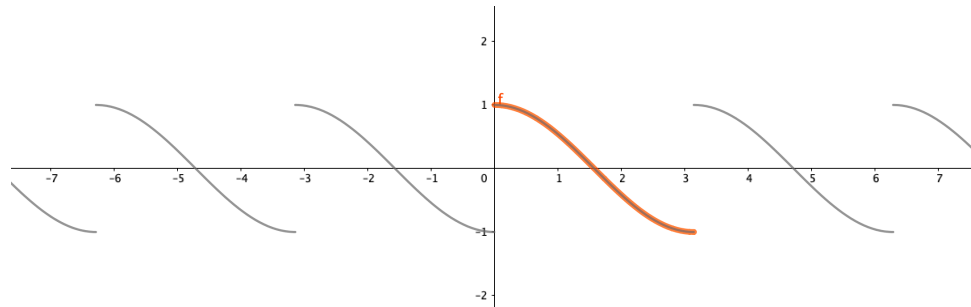


Figure 0.2: Expansion of  $f$ .

function. We check that this is the case using the definition of odd functions (Lec. 2, p. 3):

$$f(x) = -f(-x), \text{ for all } x.$$

For  $-\pi/2 < x < 0$  inputting  $-x$  in the place of  $x$  gives:

$$f(x) = -\cos x$$

$$f(-x) = -\cos(-x)$$

$$f(-x) = \cos x$$

$$f(-x) = -f(x).$$

And for  $0 < x < \pi/2$  we get

$$f(x) = \cos x$$

$$f(-x) = \cos(-x)$$

$$f(-x) = -\cos x$$

$$f(-x) = -f(x).$$

Hence, the expansion of  $f$  is odd and the corresponding Fourier series will therefore be a Fourier sine series. The period is  $p = \pi \Rightarrow L = \pi/2$ . From Lecture 2 (p. 5) the Fourier sine series is defined as

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

for

$$b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx.$$

We start by determining the Fourier coefficients  $b_n$  as

$$\begin{aligned} b_n &= \frac{4}{\pi} \int_0^{\pi/2} \cos x \sin\left(\frac{2n\pi x}{\pi}\right) dx \\ &= \frac{4}{\pi} \int_0^{\pi/2} \cos(x) \sin(2nx) dx \\ &= \frac{8n}{\pi(4n^2 - 1)}. \end{aligned}$$

Hence,

$$f(x) = \sum_{n=1}^{\infty} \frac{8n}{\pi(4n^2 - 1)} \sin(2nx).$$

b) The even expansion of  $f$  is depicted on [Figure 0.3](#). The even expansion of  $\cos x, 0 < x < \pi$  is simply  $\cos x$

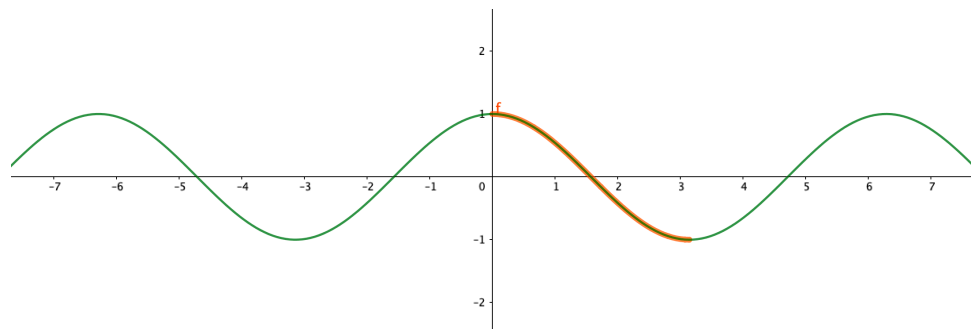


Figure 0.3: Even expansion of  $f$ .

and the Fourier series can therefore be found by comparison:

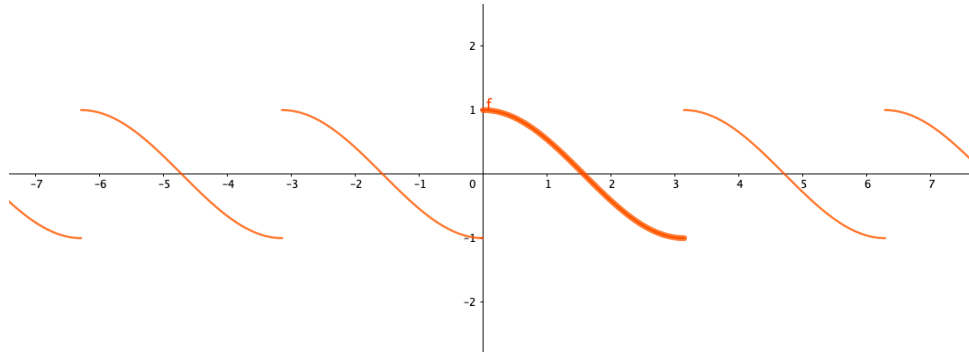
$$f(x) = \cos x$$

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right)$$

$$\Rightarrow a_0 = 0 \quad a_1 = 1.$$

c) The odd expansion of  $f$  is depicted on [Figure 0.4](#). Per definition, the odd expansion results in an odd function. Therefore, we must find a Fourier sine series. From Lecture 2 (p. 5) this is defined as

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

Figure 0.4: Odd expansion of  $f$ .

for

$$b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx.$$

We start by determining the Fourier coefficients  $b_n$  as:

$$\begin{aligned} b_n &= \frac{2}{\pi} \int_0^\pi f(x) \sin\left(\frac{n\pi x}{\pi}\right) dx \\ &= \frac{2}{\pi} \int_0^\pi \cos x \sin(nx) dx \\ &= \frac{2n}{\pi(1-n^2)} (\cos(n\pi) \cos(\pi) - 1). \end{aligned}$$

From the hints we know that this is 0 for  $n = 1$ , hence:

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right) = \sum_{n=2}^{\infty} \frac{2n}{\pi(n^2-1)} (1 - \cos(n\pi)) \sin(nx).$$

d) From the lecture (p. 5), the error  $E_N$  is defined as

$$E_N = \int_{-L}^L f^2(x) dx - L \left( 2a_0^2 + \sum_{n=1}^N (a_n^2 + b_n^2) \right).$$

For the expansion we get

$$\begin{aligned} E_N &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2(x) dx - \frac{\pi}{2} \left( \sum_{n=1}^N \left( \frac{8n}{\pi(4n^2-1)} \right)^2 \right) \\ &= \frac{\pi}{2} - \frac{\pi}{2} \left( \sum_{n=1}^N \left( \frac{64n^2}{\pi^2(4n^2-1)^2} \right) \right) \\ &= \frac{\pi}{2} - \frac{32}{\pi} \sum_{n=1}^N \frac{n^2}{(4n^2-1)^2} \\ E_1 &= \frac{\pi}{2} - \frac{32}{\pi} \cdot \frac{1}{9} = 0,439 \\ E_2 &= \frac{\pi}{2} - \frac{32}{\pi} \cdot \left( \frac{1}{9} + \frac{4}{15^2} \right) = 0,258 \\ E_3 &= \frac{\pi}{2} - \frac{32}{\pi} \cdot \left( \frac{1}{9} + \frac{4}{15^2} + \frac{9}{35^2} \right) = 0,183. \end{aligned}$$

And for the odd expansion we get:

$$\begin{aligned} E_N^{(o)} &= \int_{-\pi}^{\pi} \cos^2(x) dx - \pi \left( \sum_{n=2}^N \left( \frac{2n}{\pi(n^2-1)} (1 + \cos(n\pi)) \right)^2 \right) \\ &= \pi - \pi \left( \sum_{n=2}^N \frac{4n^2}{\pi^2(n^2-1)^2} (1 + \cos(\pi n))^2 \right) \end{aligned}$$

$$\begin{aligned}
&= \pi - \frac{4}{\pi} \left( \sum_{n=2}^N \frac{n^2}{(n^2-1)^2} (1 + \cos(\pi n))^2 \right) \\
E_1^{(o)} &= \pi - \frac{4}{\pi} \cdot 0 = \pi \\
E_2^{(o)} &= \pi - \frac{4}{\pi} \cdot \left( 0 + \frac{4}{9} \cdot 4 \right) = 0,878 \\
E_3^{(o)} &= \pi - \frac{4}{\pi} \cdot \left( 0 + \frac{16}{9} + \frac{9}{64} \cdot 0 \right) = 0,878.
\end{aligned}$$

All in all, the expansion seems to work better than the odd expansion even when taking into account the difference in interval lengths.

|                                                                                                                                                                                                                                   |        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| $ \begin{aligned} \text{a) } f(x) &= \sum_{n=1}^{\infty} \frac{8n}{\pi(4n^2-1)} \sin(2nx) \\ \text{b) } f(x) &= \cos x \\ \text{c) } f(x) &= \sum_{n=2}^{\infty} \frac{2n}{\pi(n^2-1)} (1 - \cos(n\pi)) \sin(nx). \end{aligned} $ | RESULT |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|

### Problem 3.2

Find the Fourier integral representation of

$$f(x) = \begin{cases} 1, & -1 < x < 1 \\ -1, & 1 < x < 2 \\ -1, & -2 < x < -1 \\ 0, & \text{otherwise} \end{cases}.$$

#### Derivation

From the lecture (p. 8) the Fourier integral is represented as

$$f(x) = \int_0^\infty (A(\omega) \cos(\omega x) + B(\omega) \sin(\omega x)) d\omega$$

with

$$A(\omega) = \frac{1}{\pi} \int_{-\infty}^\infty f(x) \cos(\omega x) dx$$

$$B(\omega) = \frac{1}{\pi} \int_{-\infty}^\infty f(x) \sin(\omega x) dx.$$

We determine the Fourier coefficients as:

$$\begin{aligned} A(\omega) &= \frac{1}{\pi} \left( \int_{-2}^{-1} -1 \cdot \cos(\omega x) dx + \int_{-1}^1 1 \cdot \cos(\omega x) dx + \int_1^2 -1 \cdot \cos(\omega x) dx \right) \\ &= \frac{1}{\pi} \left( \int_{-1}^1 \cos(\omega x) dx - \int_{-2}^{-1} \cos(\omega x) dx - \int_1^2 \cos(\omega x) dx \right) \\ &= \frac{4 \sin \omega - 2 \sin(2\omega)}{\omega} \\ B(\omega) &= \frac{1}{\pi} \left( \int_{-1}^1 \sin(\omega x) dx - \int_{-2}^{-1} \sin(\omega x) dx - \int_1^2 \sin(\omega x) dx \right) \\ &= 0. \end{aligned}$$

Hence, the Fourier integral representation is:

$$f(x) = \int_0^\infty \frac{4 \sin \omega - 2 \sin(2\omega)}{\omega} \cos(\omega x) d\omega$$

RESULT

### Problem 3.3

Find the Fourier integral representation of

$$f(x) = \begin{cases} -1, & -2 < x < 0 \\ 1, & 0 < x < 2 \\ 0, & \text{otherwise} \end{cases}.$$

#### Derivation

Here, we follow the same procedure as above:

$$\begin{aligned} A(\omega) &= \frac{1}{\pi} \left( \int_0^2 \cos(\omega x) \, d\omega - \int_{-2}^0 \cos(\omega x) \, d\omega \right) \\ &= 0 \\ B(\omega) &= \frac{1}{\pi} \left( \int_0^2 \sin(\omega x) \, d\omega - \int_{-2}^0 \sin(\omega x) \, d\omega \right) \\ &= \frac{1}{\pi} \cdot \frac{4 \sin^2(\omega)}{\omega} \\ &= \frac{4 \sin^2(\omega)}{\pi \omega}. \end{aligned}$$

Hence, the Fourier integral representation is:

$$f(x) = \int_0^\infty \frac{4 \sin^2(\omega)}{\omega \pi} \sin(\omega x) \, d\omega$$

RESULT

### Problem 3.4

Consider the periodic rectangular wave and its Fourier series as discussed in Lecture 1. Calculate the error of the  $N$ -th approximation for  $N = 1, 2, 3$ .

#### Derivation

From Lecture 1 pp. 5-6 the periodic rectangular wave is given by

$$f(x) = \begin{cases} -k, & -\pi < x < 0 \\ k, & 0 < x < \pi. \end{cases}$$

$$f(x) = f(x + 2\pi), \quad p = 2L = 2\pi.$$

And its Fourier coefficients are

$$a_0 = 0$$

$$a_n = 0$$

$$b_n = \frac{2k}{n\pi} (1 - \cos(n\pi)).$$

From Problem 3.1 the error  $E_N$  is defined as

$$E_N = \int_{-L}^L f^2(x) dx - L \left( 2a_0^2 + \sum_{n=1}^N (a_n^2 + b_n^2) \right).$$

Which corresponds to:

$$\begin{aligned} E_N &= \int_{-\pi}^{\pi} k^2 dx - \pi \left( \sum_{n=1}^N \left( \frac{2k}{n\pi} (1 - \cos(n\pi)) \right)^2 \right) \\ &= 2\pi k^2 - \pi \left( \sum_{n=1}^N \frac{4k^2}{n^2 \pi^2} (1 - \cos(n\pi))^2 \right) \\ &= 2\pi k^2 - \frac{4k^2}{\pi} \left( \sum_{n=1}^N \frac{1}{n^2} (1 - \cos(n\pi))^2 \right) \\ E_1 &= 2\pi k^2 - \frac{4k^2}{\pi} \cdot 4 = 1,19k^2 \\ E_2 &= E_1 \\ E_3 &= 2\pi k^2 - \frac{4k^2}{\pi} \cdot \left( 4 + \frac{4}{9} \right) = 0,624k^2. \end{aligned}$$

|                                                     |
|-----------------------------------------------------|
| $E_1 = 1,19k^2$<br>$E_2 = E_1$<br>$E_3 = 0,624k^2.$ |
|-----------------------------------------------------|

RESULT

## Lecture 4: Fourier sine and cosine transformations

19. September 2025

### Problem 4.1

Represent  $f(x)$ ,  $0 < x < \infty$ , defined as

$$f(x) = \begin{cases} \cos x, & 0 < x < \pi \\ 0, & x > \pi. \end{cases}$$

as a Fourier sine integral. *Hint:*

$$\int_0^\pi \cos x \sin(\omega x) \, dx = \frac{\omega (1 + \cos(\omega\pi))}{\omega^2 - 1}$$

### Derivation

From the Lecture p. 1 the Fourier sine integral is defined as

$$f(x) = \int_0^\infty B(\omega) \sin(\omega x) \, d\omega$$

with

$$B(\omega) = \frac{2}{\pi} \int_0^\infty f(x) \sin(\omega x) \, dx.$$

Plugging in the  $f(x)$  from the problem we get

$$\begin{aligned} B(\omega) &= \frac{2}{\pi} \int_0^\pi \cos x \sin \omega(x) \, dx \\ &= \frac{2\omega (1 + \cos(\omega\pi))}{\pi (\omega^2 - 1)}. \end{aligned}$$

Hence,

$$f(x) = \int_0^\infty \frac{2\omega (1 + \cos(\omega\pi))}{\pi (\omega^2 - 1)} \sin(\omega x) \, d\omega$$

RESULT

## Problem 4.2

Represent  $f(x)$ ,  $0 < x < \infty$ , defined as

$$f(x) = \begin{cases} \sin x, & 0 < x < \pi \\ 0, & x > \pi \end{cases}$$

as a Fourier cosine integral. *Hint:*

$$\int_0^\pi \sin x \cos(\omega x) \, dx = \frac{1 + \cos(\omega\pi)}{1 - \omega^2}.$$

## Derivation

The Fourier cosine integral is in the Lecture p. 1 given as

$$f(x) = \int_0^\infty A(\omega) \cos(\omega x) \, d\omega$$

with

$$A(\omega) = \frac{2}{\pi} \int_0^\infty f(x) \cos(\omega x) \, dx.$$

As was the case for problem 4.1 we find  $A(\omega)$  as

$$\begin{aligned} A(\omega) &= \frac{2}{\pi} \int_0^\pi \sin x \cos(\omega x) \, dx \\ &= \frac{2(1 + \cos(\omega\pi))}{\pi(1 - \omega^2)}. \end{aligned}$$

Thus,

$$f(x) = \int_0^\infty \frac{2(1 + \cos(\omega\pi))}{\pi(1 - \omega^2)} \cos(\omega x) \, d\omega$$

RESULT

### Problem 4.3

Find the Fourier cosine transform of  $f(x)$ ,  $0 < x < \infty$ , defined as

$$f(x) = \begin{cases} 1, & 0 < x < 1 \\ -1 & 1 < x < 2 \\ 0, & x > 2 \end{cases}.$$

#### Derivation

From the Lecture (p. 4) the Fourier cosine transform is given as

$$\hat{f}_C(\omega) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cos(\omega x) \, dx.$$

We find  $A(\omega)$  as

$$\begin{aligned} A(\omega) &= \sqrt{\frac{2}{\pi}} \left( \int_0^1 \cos(\omega x) \, dx - \int_1^2 \cos(\omega x) \, dx \right) \\ &= \sqrt{\frac{2}{\pi}} \frac{2 \sin(\omega) - \sin(2\omega)}{\omega}. \end{aligned}$$

Hence,

$$f(x) = \sqrt{\frac{2}{\pi}} \frac{2 \sin \omega - \sin(2\omega)}{\omega} \cos(\omega x)$$

RESULT

## Problem 4.4

Find the Fourier transform of

$$f(x) = \begin{cases} e^{2ix}, & -1 < x < 1 \\ 0, & \text{otherwise.} \end{cases}$$

### Derivation

From the Lecture (p. 7) the Fourier transform is defined as

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx.$$

By insertion we get:

$$\begin{aligned} \hat{f}(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{-1}^1 e^{2ix} e^{-i\omega x} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-1}^1 e^{i(2-\omega)x} dx \\ &= \frac{1}{\sqrt{2\pi}} \left[ \frac{e^{i(2-\omega)x}}{i(2-\omega)} \right]_{-1}^1 \\ &= \frac{1}{\sqrt{2\pi}} \left( \frac{e^{i(2-\omega)} - e^{-i(2-\omega)}}{i(2-\omega)} \right) \\ &= \frac{1}{\sqrt{2\pi}} \left( \frac{\cos(2-\omega) + i \sin(2-\omega) - \cos(\omega-2) - i \sin(\omega-2)}{i(2-\omega)} \right) \\ &= \frac{1}{\sqrt{2\pi}} \frac{2i \sin(2-\omega)}{i(2-\omega)} \\ &= \frac{1}{\sqrt{2\pi}} \frac{2 \sin(2-\omega)}{2-\omega}. \end{aligned}$$

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \frac{2 \sin(2-\omega)}{2-\omega}$$

RESULT

## Problem 4.5

Use the formulae of the Fourier transform, Fourier cosine transform and Fourier sine transform for first and second derivatives from the Lecture to derive the corresponding formulae for third derivatives, i.e. for

$$\mathcal{F}_C(f'''), \mathcal{F}_S(f'''), \mathcal{F}(f''').$$

### Derivation

From the Lecture (pp. 5 & 8) the first and second derivatives of each of the Fourier transforms are:

$$\begin{aligned}\mathcal{F}_C(f') &= \omega \mathcal{F}_S(f) - \sqrt{\frac{2}{\pi}} f(0) & \mathcal{F}_C(f'') &= -\omega^2 \mathcal{F}_C(f) - \sqrt{\frac{2}{\pi}} f'(0) \\ \mathcal{F}_S(f') &= -\omega \mathcal{F}_C(f) & \mathcal{F}_S(f'') &= -\omega^2 \mathcal{F}_S(f) + \sqrt{\frac{2}{\pi}} \omega f(0) \\ \mathcal{F}(f') &= i\omega \mathcal{F}(f) & \mathcal{F}(f'') &= -\omega^2 \mathcal{F}(f).\end{aligned}$$

Starting with the Fourier cosine transform, we will simply apply the first derivative to the second derivative as:

$$\begin{aligned}\mathcal{F}_C(f''') &= \mathcal{F}_C(f'')' \\ &= \omega \mathcal{F}_S(f'') - \sqrt{\frac{2}{\pi}} f''(0) \\ &= \omega \left( -\omega^2 \mathcal{F}_S(f) + \sqrt{\frac{2}{\pi}} \omega f(0) \right) - \sqrt{\frac{2}{\pi}} f''(0) \\ &= -\omega^3 \mathcal{F}_S(f) + \sqrt{\frac{2}{\pi}} \omega^2 f(0) - \sqrt{\frac{2}{\pi}} f''(0).\end{aligned}$$

Now we follow the same procedure for the Fourier sine and Fourier transforms as

$$\begin{aligned}\mathcal{F}_S(f''') &= \mathcal{F}_S(f'')' \\ &= -\omega \mathcal{F}_C(f'') \\ &= -\omega \left( -\omega^2 \mathcal{F}_C(f) - \sqrt{\frac{2}{\pi}} f'(0) \right) \\ &= \omega^3 \mathcal{F}_C(f) + \sqrt{\frac{2}{\pi}} \omega f'(0) \\ \mathcal{F}(f''') &= \mathcal{F}(f'')' \\ &= i\omega \mathcal{F}(f'') \\ &= i\omega (-\omega^2 \mathcal{F}(f)) \\ &= -i\omega^3 \mathcal{F}(f).\end{aligned}$$

$$\begin{aligned}\mathcal{F}_C(f''') &= -\omega^3 \mathcal{F}_S(f) + \sqrt{\frac{2}{\pi}} \omega^2 f(0) - \sqrt{\frac{2}{\pi}} f''(0) \\ \mathcal{F}_S(f''') &= \omega^3 \mathcal{F}_C(f) + \sqrt{\frac{2}{\pi}} \omega f'(0) \\ \mathcal{F}(f''') &= -i\omega^3 \mathcal{F}(f).\end{aligned}$$

RESULT

## Problem 4.6

Let

$$\hat{f}_S(\omega) = \begin{cases} 1, & 0 < \omega < 1 \\ 0, & \text{otherwise} \end{cases}.$$

Find  $f(x)$ .

### Derivation

From the Lecture (p. 5) the inverse Fourier sine transform is given as:

$$\mathcal{F}_S^{-1}(\hat{f}_S(\omega)) = f(x) = \sqrt{\frac{2}{\pi}} \int_0^\infty \hat{f}_S(\omega) \sin(\omega x) \, d\omega.$$

We substitute the  $\hat{f}_S(\omega)$  given in the assignment into the above expression to get  $f(x)$  as

$$\begin{aligned} f(x) &= \sqrt{\frac{2}{\pi}} \int_0^1 \sin(\omega x) \, d\omega \\ &= \sqrt{\frac{2}{\pi}} \frac{1 - \cos(x)}{x}. \end{aligned}$$

$$f(x) = \sqrt{\frac{2}{\pi}} \frac{1 - \cos x}{x}$$

RESULT

### Problem 4.7

Let

$$\hat{f}(\omega) = \begin{cases} 1, & -1 < \omega < 1 \\ 0, & \text{otherwise} \end{cases}.$$

Find  $f(x)$ .

### Derivation

From the Lecture (p. 7) the inverse Fourier transform is given as:

$$\mathcal{F}^{-1}(\hat{f}) = f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(\omega) e^{i\omega x} d\omega.$$

Following the same logic as in the above problem we get

$$\begin{aligned} f(x) &= \frac{1}{\sqrt{2\pi}} \int_{-1}^1 e^{i\omega x} d\omega \\ &= \frac{1}{\sqrt{2\pi}} \left[ \frac{e^{i\omega x}}{ix} \right]_{\omega=-1}^{\omega=1} \\ &= \frac{1}{\sqrt{2\pi}} \frac{e^{ix} - e^{-ix}}{ix} \\ &= \frac{1}{\sqrt{2\pi}} \frac{\cos(x) + i \sin(x) - \cos(-x) - i \sin(-x)}{ix} \\ &= \frac{1}{\sqrt{2\pi}} \frac{2 \sin(x)}{x}. \end{aligned}$$

$$f(x) = \frac{1}{\sqrt{2\pi}} \frac{2 \sin x}{x}$$

RESULT

## Problem 4.8

Consider the ODE

$$f''(x) + f(x) = r(x)$$

where

$$r(x) = \begin{cases} 1, & 0 < x < 1 \\ -1, & 1 < x < 2 \\ 0, & x > 2 \end{cases}$$

and assume that we have the boundary condition  $f'(0) = 0$ . Apply the Fourier cosine transform to both sides of the ODE to derive the corresponding equation for  $\hat{f}_C(\omega)$ . Find  $\hat{f}_C(\omega)$ .

*Hint:* You may want to use the solution of Problem 4.3.

### Derivation

We apply the Fourier cosine transform to both sides of the ODE as:

$$\mathcal{F}_C(f'' + f) = \mathcal{F}_C(r) \implies \mathcal{F}_C(f'') + \mathcal{F}_C(f) = \mathcal{F}_C(r).$$

Which corresponds to

$$\begin{aligned} -\omega^2 \hat{f}_C(\omega) + \hat{f}_C(\omega) &= \sqrt{\frac{2}{\pi}} \frac{2 \sin \omega - \sin(2\omega)}{\omega} \\ \hat{f}_C(\omega) &= \sqrt{\frac{2}{\pi}} \frac{2 \sin \omega - \sin(2\omega)}{\omega(1 - \omega^2)}. \end{aligned}$$

$$\boxed{\hat{f}_C(\omega) = \sqrt{\frac{2}{\pi}} \frac{2 \sin \omega - \sin(2\omega)}{\omega(1 - \omega^2)}}$$

RESULT

**Lecture 5: Partial Differential Equations**

26. September 2025

**Problem 5.1**

Write down the general expression for a linear first order PDE for  $u(x, y, t)$ .

---

**Derivation**

From the Lecture (p. 1) the general expression for a linear first order PDE for  $u(x, y)$  is

$$a_0(x, t) + a_1(x, t)u(x, t) + a_2(x, t)\frac{\partial u(x, t)}{\partial x} + a_3(x, t)\frac{\partial u(x, t)}{\partial t} = 0.$$

Extending this pattern we can write the general expression for a linear first order ODE for  $u(x, y, t)$  as

|                                                                                                                              |        |
|------------------------------------------------------------------------------------------------------------------------------|--------|
| $a_0(x, y, t) + a_1(x, y, t)u(x, y, t) + a_2(x, y, t)u_x(x, y, t) + a_3(x, y, t)u_y(x, y, t) + a_4(x, y, t)u_t(x, y, t) = 0$ | RESULT |
|------------------------------------------------------------------------------------------------------------------------------|--------|

## Problem 5.2

Write down the general expression for a linear second order PDE for  $u(x, y)$ . You may assume that  $u(x, y)$  is twice continuously differentiable.

---

### Derivation

Extending the logic from the previous answer, the general expression for a linear second order PDE for  $u(x, y)$  is:

$$\begin{aligned} a_0(x, y) + a_1(x, y)u(x, y) + a_2(x, y)u_x(x, y) + a_3(x, y)u_y(x, y) \\ + a_4(x, y)u_{xx}(x, y) + a_5(x, y)u_{yy}(x, y) + a_6(x, y)u_{xy}(x, y) = 0 \end{aligned}$$

RESULT

### Problem 5.3

Consider the general homogeneous first order PDE for  $u(x, t)$  with solutions  $u_1(x, t)$  and  $u_2(x, t)$ . Show that  $u(x, t) = c_1 u_1(x, t) + c_2 u_2(x, t)$  is also a solution for arbitrary constants  $c_1$  and  $c_2$ .

#### Derivation

From the Lecture (p. 2) the general homogeneous first order PDE for  $u(x, t)$  is

$$a_1(x, t)u(x, t) + a_2(x, t)u_x(x, t) + a_3(x, t)u_t(x, t) = 0.$$

We will see if this holds for arbitrary constants  $c_1$  and  $c_2$  as

$$\begin{aligned} & a_1(c_1 u_1 + c_2 u_2) + a_2 \frac{\partial c_1 u_1 + c_2 u_2}{\partial x} + a_3 \frac{\partial c_1 u_1 + c_2 u_2}{\partial t} \\ &= a_1 c_1 u_1 + a_1 c_2 u_2 + a_2 \frac{\partial c_1 u_1}{\partial x} + a_2 \frac{\partial c_2 u_2}{\partial x} + a_3 \frac{\partial c_1 u_1}{\partial t} + a_3 \frac{\partial c_2 u_2}{\partial t} \\ &= c_1 \left( a_1 u_1 + a_2 \frac{\partial u_1}{\partial x} + a_3 \frac{\partial u_1}{\partial t} \right) + c_2 \left( a_1 u_2 + a_2 \frac{\partial u_2}{\partial x} + a_3 \frac{\partial u_2}{\partial t} \right) \\ &= c_1(0) + c_2(0) \\ &= 0. \end{aligned}$$

Hence  $u(x, t) = c_1 u_1(x, t) + c_2 u_2(x, t)$  is also a solution of the general homogeneous first order PDE for  $u(x, t)$ .

## Problem 5.4

Find the solution of the one-dimensional wave equation

$$u_{tt}(x, t) = u_{xx}(x, t), \quad 0 \leq x \leq 1$$

subject to the boundary conditions

$$u(0, t) = u(1, t) = 0, \quad \text{for all } t$$

and the initial condition

$$u(x, 0) = k_0 \sin(3\pi x), \quad u_t(x, 0) = 0, \quad 0 \leq x \leq 1,$$

where  $k_0$  is a constant. You may start your calculations directly from the general solution given in the Lecture.

### Derivation

From the Lecture (p. 8) the general solution to the one-dimensional wave equation is

$$u(x, t) = \sum_{n=1}^{\infty} (B_n \cos(\lambda_n t) + B_n^* \sin(\lambda_n t)) \sin(n\pi x) \quad \lambda = n\pi.$$

From the initial elongation (p. 9) we get

$$\begin{aligned} u(x, 0) &= \sum_{n=1}^{\infty} B_n \sin(n\pi x) \\ &= k_0 \sin(3\pi x) \\ \Rightarrow B_3 &= k_0. \end{aligned}$$

And all other coefficients  $B_n$  are zero.

From the initial velocity (p. 10) we get

$$\begin{aligned} u_t(x, 0) &= \sum_{n=1}^{\infty} B_n^* \lambda_n \sin(n\pi x) = 0 \\ \Rightarrow B_n^* &= 0 \text{ for all } n. \end{aligned}$$

Hence, the particular solution is

$u(x, t) = k_0 \cos(3\pi t) \sin(3\pi x)$

RESULT

## Problem 5.5

Find the solution of the one-dimensional wave equation

$$u_{tt}(x, t) = u_{xx}(x, t), 0 \leq x \leq 1$$

subject to the boundary conditions

$$u(0, t) = u(1, t) = 0, \quad \text{for all } t$$

and the initial condition

$$u(x, 0) = k_0 \sin(3\pi x), \quad u_t(x, 0) = 3\pi k_0 \sin(3\pi x), 0 \leq x \leq 1,$$

where  $k_0$  is a constant. You may start your calculation directly from the general solution given in the Lecture.

### Derivation

As the equation, solution and initial elongation are the same as for Problem 5.4 we know that  $B_3 = k_0$ . We therefore start with the initial velocity (which differs in this case). We get

$$\begin{aligned} u_t(x, 0) &= \sum_{n=1}^{\infty} B_n^* \lambda_n \sin(n\pi x) \\ &= 3\pi k_0 \sin(3\pi x) \\ \Rightarrow B_3^* &= k_0. \end{aligned}$$

And all other coefficients  $B_n^* = 0$ . Hence,

$$u(x, t) = k_0 \sin(3\pi x) (\cos(3\pi t) + \sin(3\pi t))$$

RESULT